Duals of Hilbert Spaces; Riesz Representation Theorem; Operators in Hilbert Spaces

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Contents

0.1 Riesz Representation Theorem 2
0.2 The Adjoint 3
0.3 The Adjoint 6
0.4 Normal Operators 6
0.5 Self-adjoint operators 7
0.6 Unitary operators 8
0.7 The spectrum of an Operator 9

- The objectives are:

1-Study the adjoint of an operator
2-Study (Self-odjonit, hormal, unitary operabors)
3-Study the Spectrum of an operater.

0.1 Riesz Representation Theorem

Theorem 1 (Riesz Representation Theorem)**.** *Let H be a Hilbert space. For any bounded linear functional $\varphi \in H^*$, there exists a unique element $y \in H$ such that*

$$\varphi(x) = \langle y, x \rangle$$

for all $x \in H$ and $\|\varphi\| = \|y\|$

Now recall that $\mathbb{R}^n$, (ℓ_2 and $L_2[a,b]$ are Hilbert spaces. Using the Riesz representation theorem, we can know the nature of bounded linear functionals on these spaces as in the following examples.

Example 1. *For $F \in (\mathbb{R}^n)^*$, there exists a unique $a = (a_1, a_2, a_3, \dots, a_n) \in \mathbb{R}^n$ such that*

$$F(x) = \langle x, a \rangle = \sum_{i=1}^{n} x_i a_i, \text{ for all } x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$$

Example 2. *For $F \in (\ell_2)^*$, there exists a unique $a = (a_1, a_2, a_3, \dots, a_n, \dots) \in \ell_2$ such that*

$$F(x) = \langle x, a \rangle = \sum_{i=1}^{\infty} x_i \overline{a_i}, \text{ for all } x = (x_1, x_2, \dots, x_n, \dots) \in \ell_2$$

Example 3. *For $F \in (L_2[a,b])^*$, there exists a unique $g \in L_2[a,b]$ such that*

$$F(f) = \langle f, g \rangle = \int_a^b f(t)\overline{g}(t)dt, \text{ for all } f \in L_2$$

Corollary 1. *Every Hilbert space is reflexive.*

Corollary 2. *Let H be a Hilbert space and $T \in B(H)$, then*

$$\|T\| = \sup\{|\langle T(x), y \rangle| : x, y \in X \text{ and } \|x\| = \|y\| = 1\}.$$

Proof. Let $\alpha = \sup\{|\langle T(x), y \rangle| : x, y \in H \text{ and } \|x\| = \|y\| = 1\}$. By the Cauchy-Schwarz inequality, we have $|\langle T(x), y \rangle| \leq \|T(x)\|\|y\|$ and by the definition of the operator norm we get $\|T(x)\|\|y\| \leq \|T\|\|x\|\|y\|$. Combining these two inequalities and substituting $\|x\| = \|y\| = 1$, we obtain $|\langle T(x), y \rangle| \leq \|T\|$. Taking the supremum, we reach $\alpha \leq \|T\|$.

Now, let $x \in H$ be such that $\|x\| = 1$ and $T(x) \neq 0$. (If $T(x) = 0$ $\forall \|x\| = 1$, then $\|T\| = 0$ and so $\|T\| \leq \alpha$). As $T(x) \neq 0$, by one of the consequences of the Hahn-Banach Theorem (Corollary

6.1 from the notes), there exists $f \in H^*$ continuous such that $\|f\| = 1$ and $f(T(x)) = \|T(x)\|$. Now since $f \in H^*$ and f is bounded, by the Riesz Representation Theorem, there exists $y_f \in H$ such that $\|y_f\| = \|f\| = 1$ and $\|T(x)\| = f(T(x)) = \langle T(x), y_f \rangle$. Therefore, $\|T(x)\| \leq \alpha$ for all $\|x\| = 1$. This implies $\|T\| \leq \alpha$.

Thus, $\|T\| = \sup\{|\langle T(x), y\rangle| : x, y \in X \text{ and } \|x\| = \|y\| = 1\}$ □

Corollary 3 (The weak compactness property)**.** *Every bounded sequence in a Hilbert space H containing a weakly convergent subsequence.*

0.2 The Adjoint

Definition 1 (Adjoint Operator)**.** *Let H and G be Hilbert spaces and let $T : H \longrightarrow G$ be a bounded linear operator. An operator $T^* : G \to H$ such that*

$$\langle T(x), y\rangle_G = \langle x, T^*(y)\rangle_H \tag{1}$$

for all $x \in H, y \in G$, is called the adjoint of T.

Theorem 2. *For any bounded linear operator T on a Hilbert space H, the adjoint operator T^* exists and is unique.*

Proof. **Existence of the Adjoint:** We would like to show that for all $y \in G$, there is $w \in H$ such that

$$\langle Tx, y\rangle = \langle x, w\rangle$$

for all $x \in H$. Let

$$\begin{aligned} f_y : H &\longrightarrow \mathbb{F} \\ x &\longmapsto \langle Tx, y\rangle \end{aligned}$$

This map is linear, since it is the composition of two linear maps, namely T and $\langle \cdot, y\rangle$. Also, this map is bounded since

$$\begin{aligned} |f_y(x)| = |\langle Tx, y\rangle| &\leq \|y\|\|Tx\| \\ &\leq \|y\|\|T\|\|x\| \end{aligned}$$

So, $f_y \in H^*$. By Reisz Representation Theorem, for some $w \in H$, we can write f_y as

$$\langle Tx, y\rangle = f_y(x) = \langle x, w\rangle$$

Uniqueness of the Adjoint: Let $S : H \to G$ and $S' : H \to G$ be adjoints of T, i.e. for all $x, y \in G$

$$\begin{aligned} \langle Tx, y\rangle &= \langle x, Sy\rangle \\ \langle Tx, y\rangle &= \langle x, S'y\rangle \end{aligned}$$

We then have

$$\langle x, (S - S')y\rangle = \langle x, Sy\rangle - \langle x, S'y\rangle = \langle Tx, y\rangle - \langle Tx, y\rangle = 0$$

This implies that $(S - S')y = 0$, and hence, $S - S' = 0 \implies S = S'$. □

The defining relation of the adjoint (1) shows that the action of A on H_0 is mimicked by the action of the adjoint A^* on H_1; this mimicry is visible only through the applicable inner products, so the adjoint itself depends on these inner products. Roughly, if A has some effect, A^* preserves the geometry of that effect while acting with a reversed domain and codomain.

Theorem 3. *Let H, G be Hilbert spaces and let $T : H \to G$ and $S : H \to G$ be linear operator, then*

1. $(T^*)^* = T$
2. $(T + S)^* = T^* + S^*$
3. $(\alpha T)^* = \bar{\alpha} T^*$
4. $(TS)^* = S^* T^*$
5. $T^*(\alpha u + \beta v) = \bar{\alpha} T^*(u) + \bar{\beta} T^*(v)$

Proof.

1. We have
$$\langle (T^*)^* x, y\rangle = \overline{\langle y, (T^*)^* x\rangle} = \overline{\langle T^* y, x\rangle} = \langle x, T^* y\rangle = \langle Tx, y\rangle$$
Therefore, $(T^*)^* = T$.

2. We have
$$\begin{aligned}\langle x, (T+S)^* y\rangle = \langle (T+S)x, y\rangle &= \langle Tx, y\rangle + \langle Sx, y\rangle \\ &= \langle x, T^* y\rangle + \langle x, S^* y\rangle \\ &= \langle x, (T^* + S^*) y\rangle\end{aligned}$$
Therefore, $(T + S)^* = T^* + S^*$.

3. We have
$$\begin{aligned}\langle x, (\alpha T)^* y\rangle = \langle \alpha T x, y\rangle = \alpha \langle Tx, y\rangle = \alpha \langle x, T^* y\rangle &= \alpha \overline{\langle T^* y, x\rangle} \\ &= \overline{\bar{\alpha} \langle T^* y, x\rangle} \\ &= \overline{\langle \bar{\alpha} T^* y, x\rangle} \\ &= \langle x, \bar{\alpha} T^* y\rangle\end{aligned}$$
Therefore, $(\alpha T)^* = \bar{\alpha} T^*$.

4. We have
$$\langle x, (TS)^* y\rangle = \langle TSx, y\rangle = \langle Sx, T^* y\rangle = \langle x, S^* T^* y\rangle$$
Therefore, $(TS)^* = S^* T^*$.

5. We have
$$\begin{aligned}\langle x, T^*(\alpha u + \beta v)\rangle = \langle T(x), \alpha u + \beta v\rangle &= \overline{\langle \alpha u + \beta v, T(x)\rangle} \\ &= \bar{\alpha} \overline{\langle u, T(x)\rangle} + \bar{\beta} \overline{\langle v, T(x)\rangle} \\ &= \bar{\alpha} \langle T(x), u\rangle + \bar{\beta} \langle T(x), v\rangle \\ &= \bar{\alpha} \langle x, T^*(u)\rangle + \bar{\beta} \langle x, T^*(v)\rangle \\ &= \langle x, \bar{\alpha} T^*(u) + \bar{\beta} T^*(v)\rangle\end{aligned}$$
Therefore, $T^*(\alpha u + \beta v) = \bar{\alpha} T^* u + \bar{\beta} T^* v$.

□

Proposition 1. *If H is a complex Hilbert space and $T \in B(H)$ is invertible then T^* is invertible and $(T^*)^{-1} = (T^{-1})^*$.*

Proof. As $TT^{-1} = T^{-1}T = I$, if we take the adjoint of this equation we obtain $(TT^{-1})^* = (T^{-1}T)^* = I^*$ and so $(T^{-1})^*T^* = T^*(T^{-1})^* = I$ (see example below). Hence T^* is invertible and $(T^*)^{-1} = (T^{-1})^*$. □

Proposition 2. *Let H be a complex Hilbert space and let $T \in B(H)$, then $Ker\ T = Ker\ (T^*T)$. Moreover, the closure of $Im\ T^*$ is equal to the closure of $Im\ (T^*T)$, where $Im\ T^*$ denotes the image of T^*.*

Example 4. *Let I be the identity operator on a Hilbert space H. then $I^* = I$*

Example 5. *For $k \in C[0,1]$, let $T_k : C[0,1] \to C[0,1]$ be given by $(T_k g)(t) = k(t)g(t)$. Then $(T_k)^* = T_{\bar{k}}$.*

Example 6. *Let $T_1 : H \to H$ be given by $T_1(x) = \alpha x$ for some scalar α. For any x and y in H,*

$$\langle T_1(x), y\rangle = \langle \alpha x, y\rangle = \alpha\langle x, y\rangle = \langle x, \bar{\alpha} y\rangle$$

where the first equality follows from the definition of T_1, the second from the linearity in the first component of the inner product, and the third from the conjugate linearity in the second component of the inner product. Comparing with (1), the adjoint of T_1 is $T_1^*(y) = \bar{\alpha} y$. Hence, the adjoint of multiplication by a scalar is multiplication by the conjugate of the scalar.

From the above example it follows that if H is a real Hilbert space, then $T_1 = T_1^*$.

Example 7. *Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear operator represented by the matrix $A \in \mathbb{R}^{m\times n}$, i.e. $T(x) = Ax$. The matrix of the adjoint of T is the transpose matrix $A^T \in \mathbb{R}^{n\times m}$, since for any $x \in \mathbb{R}^m$ and $y \in \mathbb{R}^n$:*

$$\langle x, T(y)\rangle_{\mathbb{R}^m} = \langle x, Ay\rangle_{\mathbb{R}^m} = x \cdot (Ay) = x^t(Ay) = (x^tA)y = (A^tx)^ty = (A^tx)\cdot y.$$

In component notation, this identity is expressed by swapping the order of summation:

$$\sum_{i=1}^{m} x_i \left(\sum_{j=1}^{n} a_{ij} y_j\right) = \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} x_i\right) y_j.$$

For a linear map on complex spaces $T : \mathbb{C}^n \to \mathbb{C}^m$, the matrix of the adjoint with respect to the standard inner product is the Hermitian conjugate matrix, $A^* = \overline{A^T}$, satisfying $\langle x, Ay\rangle = \langle A^*x, y\rangle$.

Example 8. *Suppose that S and T are the right and left shift operators, respectively, on the sequence space $l^2(\mathbb{R})$, defined by*

$$S(x_1, x_2, x_3, \dots) = (0, x_1, x_2, x_3, \dots), \quad T(x_1, x_2, x_3, \dots) = (x_2, x_3, x_4, \dots).$$

Then $S^* = T$, since

$$\langle T(x), y\rangle = x_1(0) + x_2y_1 + x_3y_2 + x_4y_3 + \cdots = \langle x, S(y)\rangle.$$

Also, $T^* = S$, since

$$\langle S(x), y\rangle = (0)y_1 + x_1y_2 + x_2y_3 + x_3y_4 + \cdots = \langle x, T(y)\rangle.$$

Theorem 4. *Suppose that $A : \mathcal{H} \to \mathcal{H}$ is a bounded linear operator on a Hilbert space $\mathcal{H}$ with closed range. Then the equation $Ax = y$ has a solution for x if and only if y is orthogonal to* $\ker A^*$.

Example 9. *Consider the multiplication operator $M : L^2([0,1]) \to L^2([0,1])$ defined by*

$$Mf(x) = xf(x).$$

Then $M^* = M$, and M is one-to-one, so every $g \in L^2([0,1])$ is orthogonal to $\ker M^*$*; but the range of M is a proper dense subspace of $L^2([0,1])$, so $Mf = g$ is not solvable for every $g \in L^2([0,1])$.*

0.3 The Adjoint

Theorem 5. *Let H, G be Hilbert spaces, and $T : H \to G$. then*
ii) $\|T^*\| = \|T\|$
iii) $\|T^*T\| = \|T\|^2$
iv) $\ker T = (\operatorname{Im} T^*)^\perp$
v) $\ker T^* = (\operatorname{Im} T)^\perp$
vi) $\ker T^* = \{0\} \Leftrightarrow \operatorname{Im} T$ *dense in G.*

Theorem 6. *Let H be a Hilbert space and $T : H \to H$ then T is invertible $\Longleftrightarrow$* $\ker T^* = \{0\}$ *and* $\exists \alpha > 0$ *such that* $\|T(x)\| \geqslant \alpha \|x\| \quad \forall x \in H$.

Example 10. *Let $T : l^2 \to l^2$ where*

$$T(x_1, x_2 \ldots) = (0, x_1, x_2 \ldots)$$

As shown in Example 8, we have $T^*(x_1, x_2, \ldots) = (x_2, x_3, \ldots)$. So,

$$\ker T^* = \{u \in \mathbb{F} : (u, 0, 0, \ldots)\} \neq \{(0, 0, 0, \ldots)\}$$

By Theorem 6, T is not invertible.

0.4 Normal Operators

Definition 2. Let H be a Hilbert space and T a lin. operaton $T : H \to H$. then if $TT^* = T^*T, T$ is said to be normal of eratu.

Example 11. *Consider*

$$\begin{aligned} T : \mathbb{R}^2 &\longrightarrow \mathbb{R}^2 \\ x &\longmapsto \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} x \end{aligned}$$

We see that

$$T^*(x) = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}^T x = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} x$$

Since

$$\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

we have $T^*T = TT^*$.

Example 12. *If T is the right shift operator defined in Example 8, then T is not normal. This can be shown by finding T^*T and TT^*. We have*

$$T^*T(x_1, x_2, \dots) = T^*(0, x_1, x_2, \dots) = (x_1, x_2, \dots)$$

and

$$TT^*(x_1, x_2, \dots) = T(x_2, x_3, \dots) = (0, x_2, x_3, \dots)$$

which are not equal.

Example 5. If T is normal, then T - λI is also normal.

Lemma 1. *Let $T : H \to H$ be a normal operator, then*

1. $\|T(x)\| = \|T^*(x)\|\,; \quad \forall x \in H$
2. *If* $\|T(x)\| \geqslant \alpha\|x\| \quad \forall x \in H$ *then* $\ker T^* = \{0\}$. *($\alpha > 0$????)*

Proof. 1. We have

$$\|Tx\| = \langle Tx, Tx\rangle = \langle x, T^*Tx\rangle = \langle x, TT^*x\rangle = \langle T^*x, T^*x\rangle = \|T^*x\|$$

2. Let $u \in \ker T^*$, i.e. $T^*(u) = 0$. By 1, we have

$$0 = \|T^*u\| = \|Tu\| \geq \alpha\|u\| \geq 0$$

Hence, $\|u\| = 0 \implies u = 0$

□

By the lemma above and theorem 6, the next theorem immediately follows.

Theorem 7. *Let H be a Hilbert space and $T : H \to H$ is a normal operator, then T is invertible $\iff \exists\alpha >$ such thar $\quad \|T(x)\| \geqslant \alpha\|x\| \forall x \in H$.*

Hence it is easier to determine whether a normal operator invertible compared with any operator.

0.5 Self-adjoint operators

An interesting subset of the normal operators is the self-adjoint operators.

Definition 2. *$T : H \to H$ is called a self-adjoint if $T = T^*$*

Theorem 5.Let H be Hlibert space & T_1 and T_2 be self-adjont oterator, then $1 - \alpha T_1 + \beta T_2$ is a self-adjon, $\alpha\beta \in \mathbb{R}$ 2 - The set of self-odjun operuter is a clseel subset of $B(H)$.

Example 13. *For any $T \in B(H)$, we observe that TT^* and T^*T are self-adjoint, since*

$$(T^*T)^* = T^*(T^*)^* = T^*T$$

and

$$(TT^*)^* = (T^*)^*T^* = TT^*$$

Proposition 3. *Any operator $T \in B(H)$, $T = R + iS$ for R and S self-adjoint*

Proof. Let R and S be the following

$$R = \frac{T + T^*}{2} \quad S = \frac{T - T^*}{2i}$$

then we have

$$R + iS = \Big(\frac{T + T^*}{2}\Big) + i\Big(\frac{T - T^*}{2i}\Big) = \frac{T + T^*}{2} + \frac{T - T^*}{2} = \frac{2T}{2} = T$$

and

$$R^* = \Big(\frac{T + T^*}{2}\Big)^* = \frac{(T + T^*)^*}{2} = \frac{T^* + (T^*)^*}{2} = \frac{T^* + T}{2} = \frac{T + T^*}{2} = R$$

$$S^* = \Big(\frac{T - T^*}{2i}\Big)^* = -\frac{(T - T^*)^*}{2i} = -\frac{T^* - (T^*)^*}{2} = -\frac{T^* - T}{2} = \frac{T - T^*}{2} = S$$

□

0.6 Unitary operators

Definition 3. *An operator $T \in B(H)$ called unitary, if $TT^* = T^*T = I$.*

Example 14. *It can be see that the identity operator, I, is unitary, since $I^*I = II = I = II = II^*$.*

Lemma.If $S, T \in B(H)$ and
$\langle Sz, z\rangle = \langle Tz, 3\rangle \quad \forall z \in H$
$\Rightarrow \quad S = T$.

Theorem 6.Let $T \in B(H)$, then T is unitary if T is isometry.

Theorem 8. *Let T, S be unitary operators, then*

1. *T^* is unitary and $\|T\| = \|T^*\| = 1$*
2. *TS is unitary*
3. *T^{-1} is unitary.*

Remark 1. *By Theorem 8, Example 14 and the associativity of function composition, we observe that the set of all unitary operators forms a group under composition. Moreover, if $T : \mathbb{C}^n \to \mathbb{C}^n$, this group is called the **unitary group**, denoted $U(n)$. There is a special subgroup of $\mathrm{U}(n)$, which is the **special unitary group**, denoted $\mathrm{SU}(n)$, which is the set of all unitary operators from $\mathbb{C}^n$ to $\mathbb{C}^n$ such that its determinant is 1. This forms a subgroup since if we let $A, B \in \mathrm{SU}(n)$, we have*

$$\det AB^{-1} = \frac{\det A}{\det B} = \frac{1}{1} = 1$$

showing that $AB^{-1} \in \mathrm{SU}(n)$.

0.7 The spectrum of an Operator

The notation of an eigenvalue of a matrix has many essential applications in finite dimensional vector space; however, we wish we could extend this notation to the infinite-dimensional vector space.

Definition 4. *Let $T \in B(H), \lambda \in \mathbb{C}$ is called an eigenvalue of T, if there exsits $x \in H$ such that*

$$Tx = \lambda x$$

Definition 5. *Let $T \in B(H)$. The spectrum of T, denoted by $\sigma(T)$ is*
$\sigma(T) = \{\lambda \in \mathbb{C} : T - \lambda I \text{ is not invertible } \}$

Theorem 9. *If λ is an eigenvalue of T , then $\lambda \in \sigma(T)$*

Remark for finite dimesonial Hilbert space, the spectrum of T cosist only the eigenvalues. However, in Infite dimesond Hilbert Spare we may have no eigenvalus.

Example: Consider $S \in B\left(l^2\right)$ $S\left(x_1, x_2, \ldots\right) = (0, x_1, x_2, \ldots)$ has no egienvalue

Example 15. *Let $\mathcal{H}$ be a complex Hilbert space and let $T \in B(\mathcal{H})$.*
If $|\lambda| > \|T\|$ then $\lambda \notin \sigma(T)$.

Lemma 2. *If $\mathcal{H}$ is a complex Hilbert space and $T \in B(\mathcal{H})$ then $\sigma\left(T^*\right) = \{\bar{\lambda} : \lambda \in \sigma(T)\}$.*

Example 16. *If $S : \ell^2 \to \ell^2$ is the unilateral shift then:*
(a) if $\lambda \in \mathbb{C}$ with $|\lambda| < 1$ then λ is an eigenvalue of S^*;
(b) $\sigma(S) = \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$.

Theorem 10. *Let $\mathcal{H}$ be a complex Hilbert space and let $T \in B(\mathcal{H})$.*
If T is invertible then $\sigma\left(T^{-1}\right) = \left\{\mu^{-1} : \mu \in \sigma(T)\right\}$.

Lemma 3. *If $\mathcal{H}$ is a complex Hilbert space and $U \in B(\mathcal{H})$ is unitary then*

$$\sigma(U) \subseteq \{\lambda \in \mathbb{C} : |\lambda| = 1\}$$

Corollary 4. *If A is a self-adjoint matrix with eigenvalues $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$, then*

$$\|A\| = \max\left\{|\lambda_1|, |\lambda_2|, \ldots, |\lambda_n|\right\}.$$